

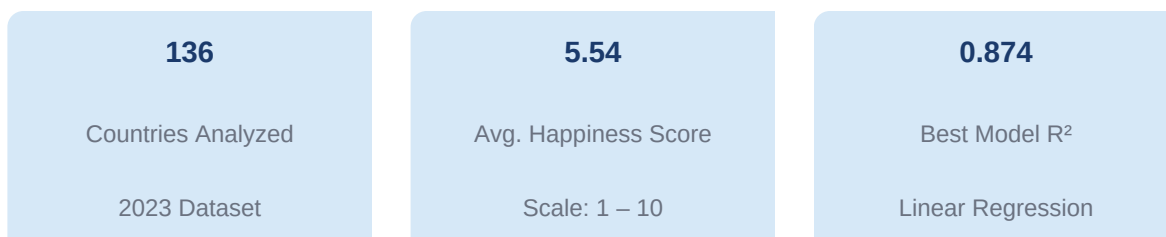


Table of Contents

01	Executive Summary Key findings at a glance
02	Data Overview Dataset description & statistics
03	The World's Happiest Nations Top 10 rankings & insights
04	GDP & Happiness Does wealth equal well-being?
05	Factor Analysis Correlation heatmap deep-dive
06	Distribution & Anomalies Global spread & over-performers
07	Predictive Modeling Three ML models compared
08	Conclusions & Recommendations Actionable insights for leaders

Executive Summary

This analysis of the 2023 World Happiness Report explores the primary drivers of national happiness across **136 countries**. Using statistical analysis and machine learning, we identify what truly makes nations thrive — and the results challenge conventional assumptions about the primacy of economic wealth.



Key Findings

- **GDP is foundational, but not the ceiling.** Logged GDP per capita correlates strongly (~0.78) with happiness, yet nations with equal wealth show very different well-being outcomes.
- **Social support matters as much as wealth.** Countries with strong social safety nets consistently outperform economically similar peers.
- **Health = happiness.** Healthy life expectancy tracks nearly as closely with happiness as GDP does.
- **Freedom amplifies well-being.** Freedom to make life choices is a key differentiator among mid-tier economies.
- **Cultural factors defy models.** Venezuela, Congo, and Honduras score far higher than economic metrics predict.

BOTTOM LINE: Sustainable national happiness requires investing in social infrastructure, public health, and individual freedoms — not just economic growth.

Data Overview

The dataset is drawn from the **2023 World Happiness Report**, which surveys citizens across 136 nations. After removing rows with missing values, the final clean dataset contains complete records used for all analyses and modeling.

Happiness Score (Ladder Score) — Summary Statistics

Metric	Value
Count (countries)	136
Mean Happiness Score	5.54
Std Deviation	1.14
Minimum	1.86 (most unhappy)
25th Percentile (Q1)	4.70
Median (Q2)	5.69
75th Percentile (Q3)	6.34
Maximum	7.80 (happiest)

Features Used in Analysis

Feature	Description
Logged GDP per capita	Economic output per person (log-transformed)
Social support	Having someone to count on in times of trouble
Healthy life expectancy	Average years of healthy life from birth
Freedom to make choices	Satisfaction with freedom of life choices
Generosity	Recent donations relative to national income
Perceptions of corruption	Absence of corruption in govt. & business



Top 10 Happiest Countries (2023)

Finland, Denmark, and Iceland lead the 2023 rankings. Nordic nations dominate the top tier, reflecting strong social systems, high trust, and excellent public health outcomes.

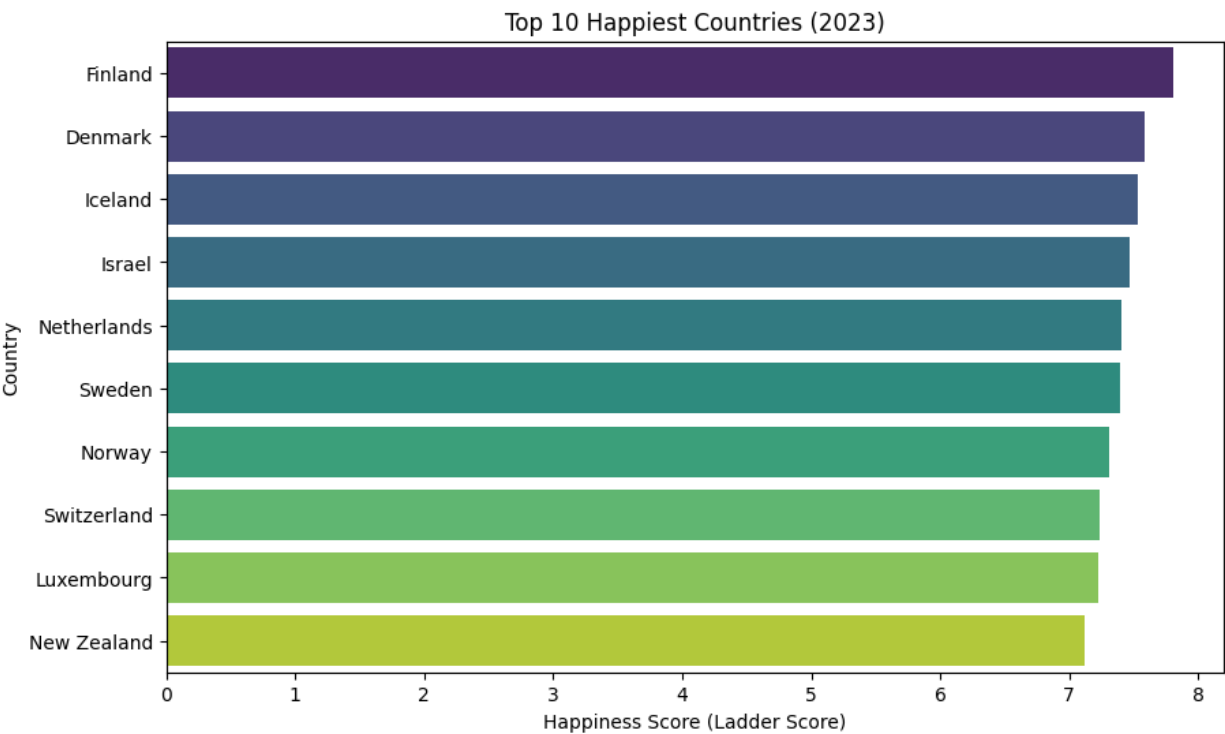


Figure 1 · Top 10 Happiest Countries by Ladder Score

What the Rankings Tell Us

- **Nordic dominance:** Finland, Denmark, Iceland, Sweden, and Norway all appear in the top 10. Their shared traits: universal healthcare, high social trust, and strong safety nets.
- **Israel at #4:** Despite geopolitical tensions, Israel scores 7.47 — pointing to strong community bonds and quality of life.
- **New Zealand & Australia** represent the only non-European nations in the top tier, supported by high GDP, freedom scores, and life expectancy.



GDP per Capita vs. Happiness Score

The scatter plot below shows the relationship between a country's logged GDP per capita and its Ladder happiness score. A regression line reveals the trend — but the spread around it tells an equally important story.



Figure 2 · Logged GDP per Capita vs. Ladder Score with Regression Line

Interpretation for Leaders

Strong Positive Trend	Wealthier nations are generally happier — GDP correlation with happiness is approximately 0.78 .
Wide Variance Exists	Countries at the same GDP level show wildly different happiness scores — meaning wealth alone cannot explain national well-being.
Diminishing Returns	At higher GDP levels, additional wealth yields smaller happiness gains. Social and governance factors take over.



Correlation Between Happiness Factors

The heatmap below displays pairwise correlations between all key variables. Values closer to **+1.0** indicate strong positive relationships; values near **0** suggest no link.

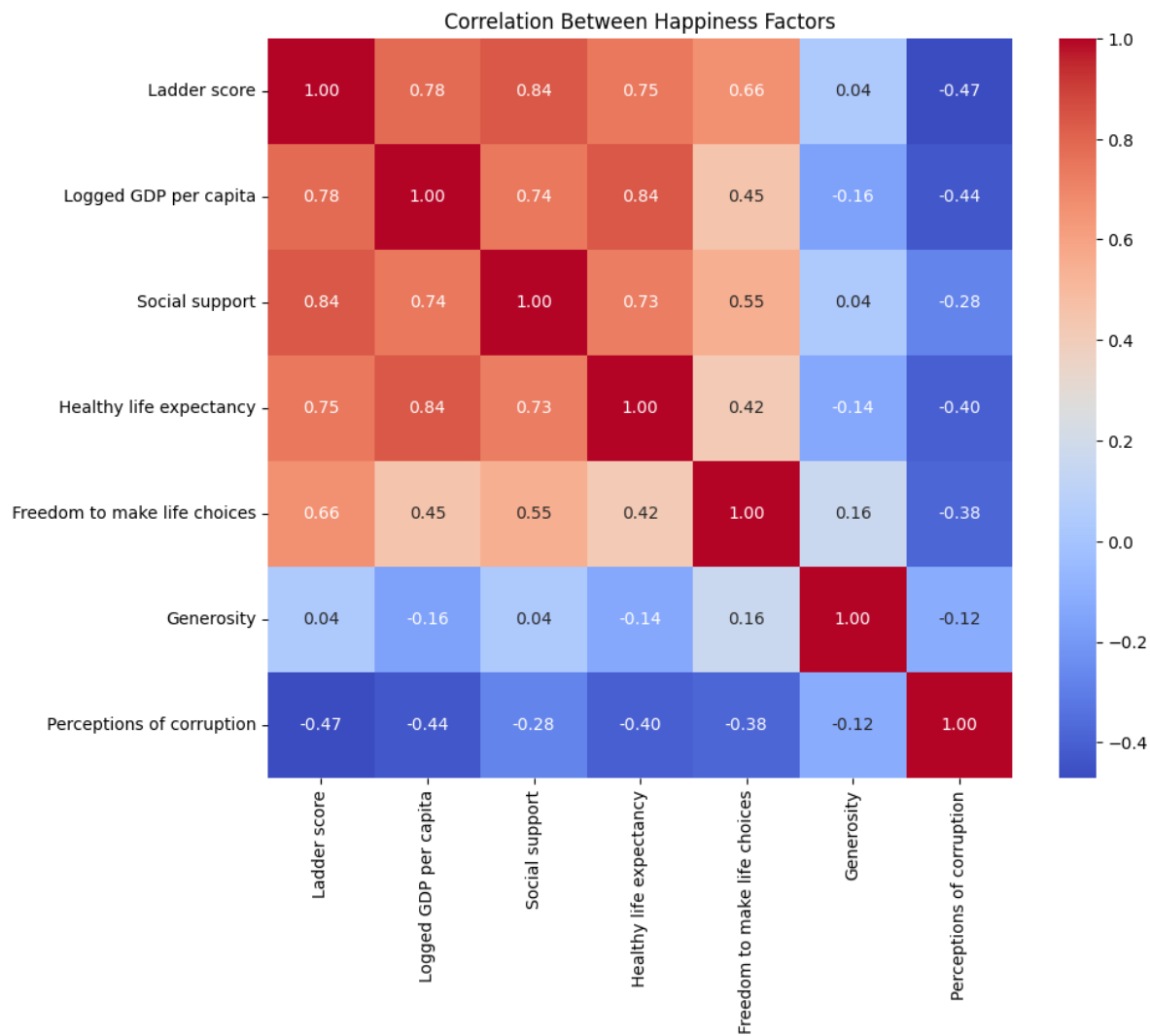


Figure 3 · Correlation Heatmap of All Happiness Factors

Top Correlations with Happiness (Ladder Score)

Factor	Correlation	Insight
Logged GDP per capita	~0.78	Strongest single predictor
Social support	~0.75	Nearly as powerful as wealth
Healthy life expectancy	~0.74	Health drives happiness globally

Freedom to make life choices	~-0.55	Key in mid-tier economies
Generosity	~-0.17	Weak direct link; cultural factor
Perceptions of corruption	~-0.45	More corruption = less happiness



Global Distribution of Happiness Scores

The histogram below shows how happiness scores are distributed across 136 countries. A slight **left (negative) skew of -0.46** indicates that most countries cluster around the middle-to-upper range, with fewer very unhappy nations dragging the tail.



Figure 4 · Distribution of Global Happiness Scores (KDE overlay)

Countries Happier Than Predicted ('Over-Performers')

By examining the *Dystopia + residual* column, we identify nations that score significantly higher than their economic and social metrics would predict. These are the most intriguing anomalies in the dataset.

Rank	Country	Ladder Score	Residual	Insight
1	Venezuela	5.21	+2.955	Cultural resilience despite economic crisis
2	Congo (Brazzaville)	5.27	+2.802	Community bonds outperform metrics
3	Guinea	5.07	+2.705	Strong social fabric & family ties
4	Israel	7.47	+2.691	Already #4 globally — further exceeds model
5	Honduras	6.02	+2.631	High happiness despite poverty indicators

INSIGHT: Cultural identity, community solidarity, and shared purpose can generate happiness that economic models cannot capture or predict.



Machine Learning Models: Predicting Happiness

Three regression models were trained on an 80/20 train-test split (**109 training** / **28 test** countries) using the six happiness factors as features.

Model Performance Comparison

Model	MAE ↓ (lower = better)	R ² ↑ (higher = better)	Verdict
Linear Regression	0.284	0.874	■ Best Overall
Random Forest	0.291	0.841	■ Good — interpretable
Gradient Boosting	0.308	0.810	■ Solid but overfits slightly

Actual vs. Predicted Happiness (Random Forest)

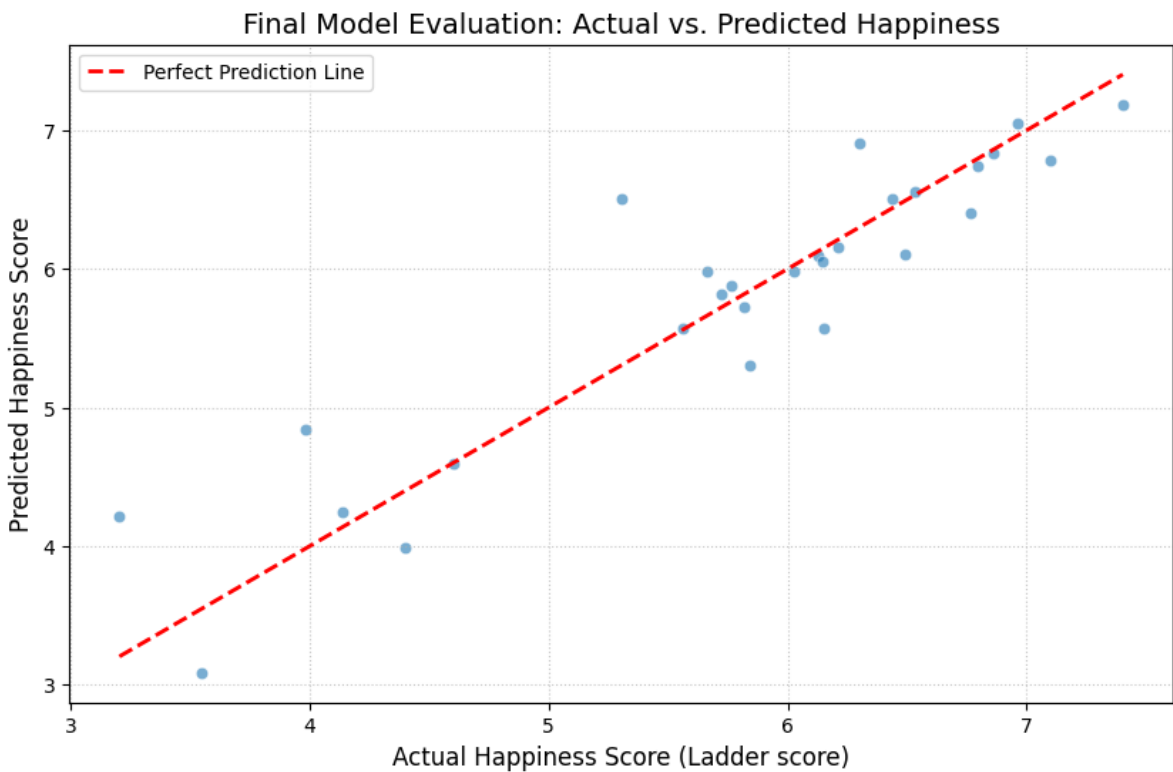


Figure 5 · Actual vs. Predicted Happiness Scores — points near the red line indicate accurate predictions

Feature Importance (Random Forest)

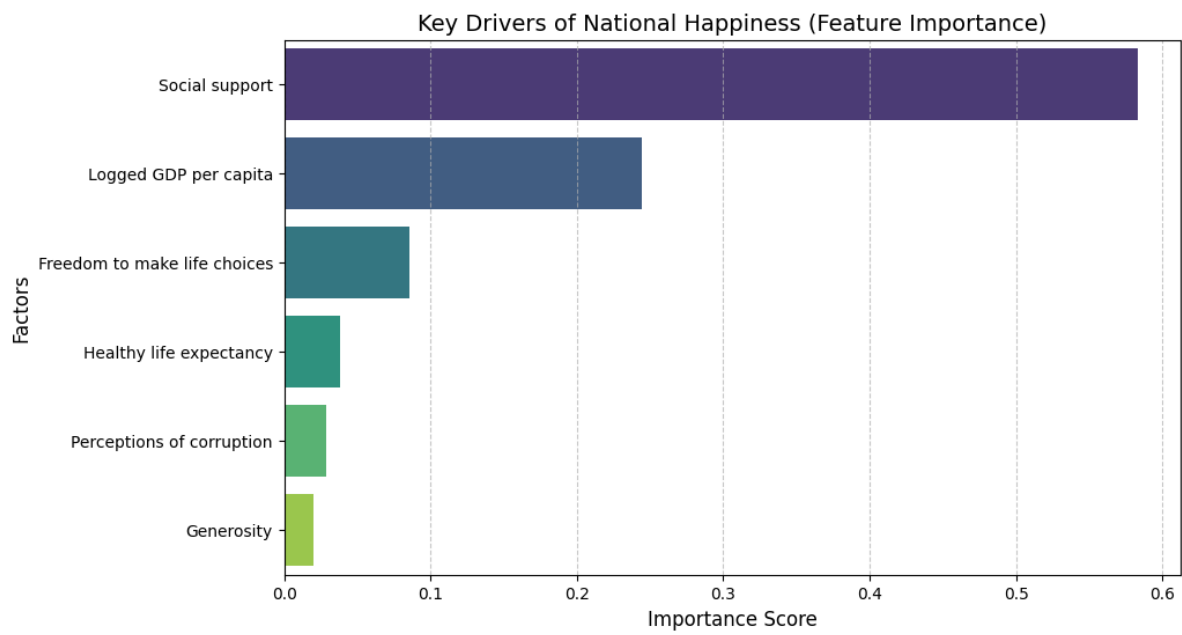


Figure 6 · Key Drivers of National Happiness by Importance Score



Conclusions & Recommendations

The data is clear: **economic growth alone cannot guarantee a happy nation.** The following recommendations are grounded in the statistical analysis and model results.

Invest in Social Safety Nets

Social support is nearly as predictive of happiness as GDP. Universal healthcare, unemployment benefits, and strong community programs deliver measurable happiness returns.

Prioritize Public Health

Healthy life expectancy is the third-strongest happiness predictor. Investments in preventive healthcare and wellness programs are both economically and socially wise.

Protect Individual Freedoms

Freedom to make life choices is critical for mid-tier economies. Reducing bureaucratic barriers and protecting civil liberties amplifies national well-being.

Combat Corruption Decisively

Corruption has a significant negative correlation with happiness. Transparent governance and accountability institutions are essential for sustained well-being.

Study Cultural Over-Performers

Countries like Venezuela and Honduras beat economic predictions. Understanding the cultural and social factors behind this can yield policy insights not captured by GDP models.

A happy nation is built on three pillars: Economic Prosperity · Social Solidarity · Personal Freedom. None of these pillars stands strong alone.

Data Source: 2023 World Happiness Report · Analysis performed in Python (pandas, seaborn, scikit-learn) · 136 countries included